

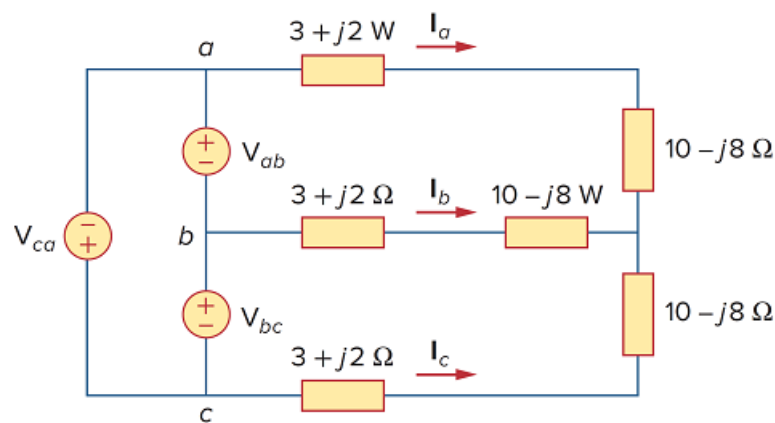
Problem

In the circuit of Fig. 12.54, if $\mathbf{V}_{ab} = 440\angle 10^\circ$,

$\mathbf{V}_{bc} = 440\angle -110^\circ$, $\mathbf{V}_{ca} = 440\angle 130^\circ$ V, find the

line currents.

Figure 12.54



Step-by-step solution

Step 1 of 1

Convert the delta connected source to an equivalent WYE – connected source and consider the single phase equivalent.

$$I_a = \frac{440\angle(10^\circ - 30^\circ)}{\sqrt{3}Z_y}$$

Where,

$$Z_y = 3 + j2 + 10 - j8 = 13 - j6 = 14.32\angle -24.78^\circ$$

$$I_a = \frac{440\angle -20^\circ}{\sqrt{3}(14.32\angle -24.78^\circ)} = 17.74\angle 4.78^\circ \text{ A}$$

$$I_b = I_a\angle -120^\circ = 17.74\angle -115.22^\circ \text{ A}$$

$$I_c = I_a\angle 120^\circ = 17.74\angle 124.78^\circ \text{ A}$$